QUIZ B

Q. Consider a library's database that includes two tables:

- ✓ **Books** with book id (Primary Key), title (Not Null), author (Not Null), publication_year and genre.
- ✓ **Borrowing** with borrow_id (Primary Key), book_id (Foreign Key referencing books table book_id), borrower_name (Not Null), borrow_date (Default to the current date) and return_date (Nullable).

Insert the following data in both the tables:

BOOKS

(101, 'Dune', 'Frank Herbert', 1965, 'Science Fiction')
(102, 'The Hobbit', 'J.R.R. Tolkien', 1937, 'Fantasy')
(103, 'The Da Vinci Code', 'Dan Brown', 2003, 'Mystery')
(104, 'Sapiens', 'Yuval Noah Harari', 2011, 'Historical')
(105, 'Neuromancer', 'William Gibson', 1984, 'Science Fiction')

BORROWING

(201, 101, 'Alice Smith', '2024-08-01', NULL)
(202, 102, 'Bob Johnson', '2024-08-05', '2024-08-15')
(203, 103, 'Carol White', '2024-08-10', '2024-08-20')
(204, 104, 'David Brown', '2024-08-12', NULL)
(205, 105, 'Eve Davis', '2024-08-15', '2024-08-25');

The library needs you to perform the following tasks:

1. Write a query to list the titles and publication years of all books in the "Science Fiction" genre. The results should be ordered by publication year in ascending order.
2. Update the genre of the book with book_id 101 to "Fantasy". Ensure that the genre is updated only if the book's publication year is before 2000.
3. Insert a new borrowing record with book_id 202, borrower name "John Doe", and today's date. Ensure the record is inserted only if the book_id exists in the Books table.
4. Delete all borrowing records where the return_date is not yet set (i.e., still NULL).
5. Write a query to find the titles of books that are borrowed by more than 2 different people. Use a subquery to first find book_ids with more than 2 distinct borrowers, and then retrieve the titles of those books.
6. Write a query to find the average publication year of books in the "Historical" genre. Ensure that only books published after 1900 are considered. The result should be grouped by genre.
7. Describe how you would add a CHECK constraint to ensure that the publication_year column in the Books table only accepts values from 1900 onwards.